

Introduce one means to DSE Backup and Recovery

Discussion Unit:

DSE Core Backup and Recovery



Discussion Lab:

List Why ? (You might restore when _____ ?)



Source: www.dilbert.com

List Use Cases: You might restore when _____ ?



Source: <https://www.pinterest.com/pin/16958936066982309/>

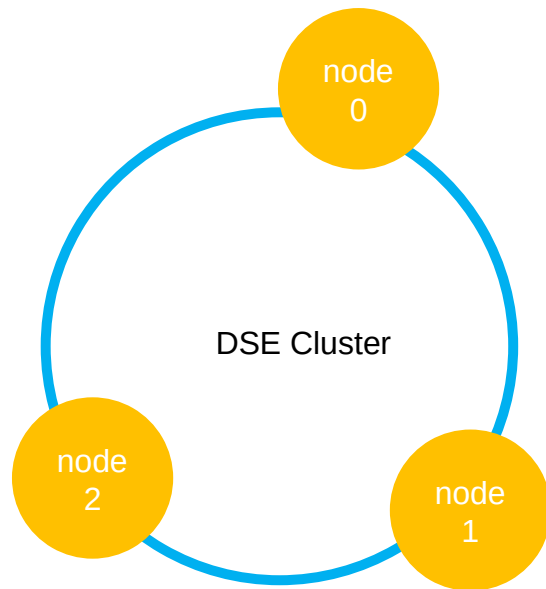
- 1.
- 2.
- 3.
- 4.



End of Discussion Lab:

DSE: Backup and Restore

- Ops Center Backup Server, including to S3
- sstabledump
- snapshots
- Other



And what about consistency on restore ?

Review of Runtime-

```
cassandra.yaml
  data_file_directories:
    - /opt/dse/node1/data/data
```

```
cd /opt/dse/node1/data/data
ls -l
```



What are these ?

```
drwxr-xr-x  4 root root 4096 Jul  1 11:59 dse_leases
drwxr-xr-x  3 root root 4096 Jul  1 11:59 dse_perf
drwxr-xr-x  5 root root 4096 Jul  1 11:58 dse_security
drwxr-xr-x  3 root root 4096 Jul  1 11:59 dse_system
drwxr-xr-x  3 root root 4096 Jul  1 11:59 dse_system_local
drwxr-xr-x  3 root root 4096 Jul  1 11:59 solr_admin
drwxr-xr-x 18 root root 4096 Jul  1 11:58 system
drwxr-xr-x  5 root root 4096 Jul  1 11:58 system_auth
drwxr-xr-x  7 root root 4096 Jul  1 11:58 system_distributed
drwxr-xr-x 12 root root 4096 Jul  1 11:58 system_schema
drwxr-xr-x  4 root root 4096 Jul  1 11:58 system_traces
```

Make a Keyspace/Table (cust_orders) ..

```
...  
drwxr-xr-x  3 root root 4096 Jul  1 11:59 dse_system_local  
drwxr-xr-x  3 root root 4096 Jul  1 11:59 ks_6254  
drwxr-xr-x  3 root root 4096 Jul  1 11:59 solr_admin  
...
```

```
cd ks_6254  
ls -l  
drwxr-xr-x 3 root root 4096 Jul  1 11:59 cust_orders-8eda10 ...
```

```
cd cust*  
ls -l  
drwxr-xr-x 2 root root 4096 Jul  1 11:59 backups
```

```
cd back*  
ls -l
```

(All above just empty directories)



Why the
UUID ?

Take a snapshot

```
nodetool cleanup ks_6254
```

```
nodetool snapshot -t ks_6254-snap1 ks_6254
```

Requested creating snapshot(s) for [ks_6254] with
snapshot name [ks_6254-snap1] and options {skipFlush=false}
Snapshot directory: ks_6254-snap1

Contents of keyspace data directory

```
pwd
```

```
/opt/dse/node1/data/data/ks_6254/cust_orders-  
8eda10d17d5811e8839b2baf872cd95b
```

```
ls -l
```

```
-rw-r--r-- 2 root root  47 Jul  1 12:04 aa-1-bti-CompressionInfo.db  
-rw-r--r-- 2 root root 135 Jul  1 12:04 aa-1-bti-Data.db  
-rw-r--r-- 2 root root  10 Jul  1 12:04 aa-1-bti-Digest.crc32  
-rw-r--r-- 2 root root  16 Jul  1 12:04 aa-1-bti-Filter.db  
-rw-r--r-- 2 root root  73 Jul  1 12:04 aa-1-bti-Partitions.db  
-rw-r--r-- 2 root root   0 Jul  1 12:04 aa-1-bti-Rows.db  
-rw-r--r-- 2 root root 4782 Jul  1 12:04 aa-1-bti-Statistics.db  
-rw-r--r-- 2 root root  94 Jul  1 12:04 aa-1-bti-TOC.txt  
drwxr-xr-x 2 root root 4096 Jul  1 11:59 backups  
drwxr-xr-x 3 root root 4096 Jul  1 12:04 snapshots
```

Contents of snapshot directory

```
cd snapshots
```

```
ls -l
```

```
drwxr-xr-x 2 root root 4096 Jul  1 12:04 ks_6254-snap1
```

```
cd ks*
```

```
ls -l
```

```
-rw-r--r-- 2 root root  47 Jul  1 12:04 aa-1-bti-CompressionInfo.db
-rw-r--r-- 2 root root 135 Jul  1 12:04 aa-1-bti-Data.db
-rw-r--r-- 2 root root  10 Jul  1 12:04 aa-1-bti-Digest.crc32
-rw-r--r-- 2 root root  16 Jul  1 12:04 aa-1-bti-Filter.db
-rw-r--r-- 2 root root  73 Jul  1 12:04 aa-1-bti-Partitions.db
-rw-r--r-- 2 root root   0 Jul  1 12:04 aa-1-bti-Rows.db
-rw-r--r-- 2 root root 4782 Jul  1 12:04 aa-1-bti-Statistics.db
-rw-r--r-- 2 root root  94 Jul  1 12:04 aa-1-bti-TOC.txt
-rw-r--r-- 1 root root  31 Jul  1 12:04 manifest.json
-rw-r--r-- 1 root root 942 Jul  1 12:04 schema.cql
```

Truncate table ..

```
cqlsh> truncate table ks_6254.cust_orders;  
cqlsh> selec * from ks_6254.cust_orders;  
(0 rows)
```

```
pwd  
/opt/dse/node1/data/data/ks_6254/cust_orders- \8eda10d17d5811e8839b2baf872cd95b/snapshots
```

```
ls -l  
drwxr-xr-x 2 root root 4096 Jul  1 12:04 ks_6254-snap1  
drwxr-xr-x 2 root root 4096 Jul  1 12:11 truncated-1530468681275-cust_orders
```

nodetool clearsnapshot --all
Or, snapshot name

(Don't do this now.)



What is
this ?

From snapshot can ..

Can restore to new cluster; see Url on Notes page

Can restore to original cluster,

cp

```
/opt/dse/node1/data/data/ks_6254/cust_orders-  
8eda10d17d5811e8839b2baf872cd95b/snapshots/ks_6254-snap1/* \  
/opt/dse/node1/data/data/ks_6254/cust_orders-8eda10d17d5811e8839b2baf872cd95b
```

cp source destination

cp snapshot_dir / data_dir*

nodetool refresh, Check results

```
nodetool refresh ks_6254 cust_orders
```

```
cqlsh:ks_6254> select * from cust_orders;
```

region	cust_name	ord_num	other
NA	SEARS	101	Shoes, Washer
NA	SEARS	102	Oranges
EMEA	IKEA	101	Shoes
NA	MACYS	101	Dress, Tie

(4 rows)

What about multi-node ?

Same exact procedure, run on each node concurrently

By the nature of the snapshots (each taken milliseconds/seconds apart), data is restored to 'eventual consistency'

DSE runtime will create a consistent result

